

Dart Part 2

DMIT 2504 WEEK 1 DAY 3



Announcements

Expectations

- ▶ No Late Assignments
- ▶ No Cheating
- ▶ Be a good classmate
- ▶ Don't go to stupid sites in class
- ▶ Show up to class

Review

- ▶ Creating a dart project
- ▶ lib and bin
- ▶ Input and output
- ▶ Classes / Objects
- ▶ Exceptions

Functions

Return Type

Name (follow proper style guidelines)

```
/// Returns a [String]
///
/// Displays the passed in [message] as the prompt
/// for the user.
String prompt(String message) {
    stdout.write(message);
    return stdin.readLineSync()!;
}
```

Parameters

Some Notes on Function Parameters

- ▶ By default parameters are ordered and required
 - ▶ `void doSomething(String param1, int param2){...}`
- ▶ Optional positional parameters can be added to the end with []
 - ▶ `void doSomething(String param1, [int? param2]){...}`
- ▶ Default values can be assigned with =
 - ▶ `void doSomething(String param1, int param2 = 10) {...}`
- ▶ Parameters can be named by using {}.
 - ▶ `void doSomething({String? param1})`
 - ▶ `doSomething(param1: "Ice Cream")`
- ▶ Optional positional parameters need to be nullable (?) or have a default value. More on null safety to follow
- ▶ Named parameters need to be nullable, required, or have a default value

Lists

```
List dynamicList = [];  
  
print(dynamicList.length); // Prints 0  
  
dynamicList.add("Hello");  
  
print(dynamicList[0]); // Prints "World"  
  
print(dynamicList.length); // Prints 1  
  
List preFilledDynamicList = [1, 2, 3];  
  
print(preFilledDynamicList[0]); // Prints 1  
  
print(preFilledDynamicList.length); // Prints 3  
  
dynamicList += preFilledDynamicList;  
  
print(dynamicList); // Prints [Hello, 1, 2, 3]  
  
List<String> stringList = [];  
  
stringList.add("I'm a string");  
  
stringList.add(1); Won't Work
```

Null Safety

- ▶ Variables in Dart, by default, cannot be null
- ▶ To allow something to be null you add a ?
- ▶ If something will be assigned before it is accessed, you can add the late keyword
- ▶ To check for nullable variables existence, you can use `if (variable != null)` or `?.`

```
String? goalScorer;  
// Other code  
if (goalScorer != null) {  
  print(goalScorer.length);  
}
```

```
String? goalScorer;  
// Other code  
// ignore: dead_code  
print(goalScorer?.length);
```

- ▶ You can also tell the compiler that something is not null with `!`. This is dangerous and should only be used when confident

```
String? goalScorer;  
// Other code  
print(goalScorer!.length);
```